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Malaria Transmission and Insecticide Resistance of *Anopheles gambiae* (Diptera: Culicidae) in the French Military Camp of Port-Bouët, Abidjan (Côte d'Ivoire): Implications for Vector Control

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ABSTRACT An important vector control program is ongoing to lower the risk of malaria transmission in the French military camp of Port-Bouët, Abidjan (Côte d'Ivoire). However, some autochthonous malaria cases are regularly suspected. An entomological survey was conducted in June 2004 in the camp to assess malaria transmission and evaluate the pyrethroid and organophosphate resistance of the malaria vectors. The average mosquito biting rate was 178.0 bites per person per night. Mosquitoes belonging to the *Anopheles gambiae* (Diptera: Culicidae) complex and the *Anopheles funestus* group were collected. *An. gambiae* s.s. molecular form M was the only species of the *An. gambiae* complex present. The average number of *An. gambiae* bites was ≈44.3 per person per night. The circumsporozoite index was 0.38% and the entomological inoculation rate estimated to be 1.2 infective bites per week for the study period. The *kdr* and *ace1* gene frequencies in the *An. gambiae* population were 0.70 and 0.15, respectively. Personnel living in the French barracks of Port-Bouët are thus at high risk of being bitten by parasite-infected mosquitoes. Such an entomological inoculation rate, usually found in African peri-urban environments, was unexpected considering the extensive effort deployed to control mosquitoes in the camp. Insecticide resistance could explain the inefficacy of the vector control program but the spraying strategy is also questionable.

KEY WORDS *Anopheles gambiae*, insecticide resistance, malaria, Ivory Coast

Malaria remains a major endemic disease in tropical areas, particularly in Africa, which pays a heavy price every year in terms of morbidity and mortality, with an estimated 200–300 million clinical cases and ≈1 million deaths annually because of *Plasmodium falciparum* (WHO 2005). In Côte d'Ivoire, malaria is the chief reason for admission in hospitals and the main cause of mortality recorded in rural sanitary services, affecting mainly children (Dossou-Yovo et al. 1998). This situation is exacerbated by the emerging resistance of *P. falciparum* to commonly used drugs (Djaman et al. 2002, Assi et al. 2004) and resistance in vector mosquitoes to the different classes of insecticides used for vector control (Elissa et al. 1993, N'Guessan et al. 2003). Malaria transmission is essentially through members of the *Anopheles gambiae* (Diptera: Culicidae) complex and *Anopheles funestus* (Escudie et al. 1962, Coz et al. 1966, Dossou-Yovo et al. 1995, Henry et al. 2003). Three species of the *An. gambiae* complex are present in Côte d'Ivoire: *Anopheles melas*, *Anopheles arabiensis*, and *An. gambiae* s.s., the most efficient malaria vector in West Africa (Gillies and Coetzee 1987). Recent research programs have revealed the existence of two genetic variants in *An. gambiae* s.s. now referred to as M and S forms with strong evidence of reproductive isolation (Wondji et al. 2002). Both molecular forms have been detected in Côte d'Ivoire (Chandre et al. 1999b, Della Torre et al. 2001). The *kdr* and *ace1* genetic mutations, conferring cross-resistance to pyrethroids/DDT and carbamate/organophosphate insecticides, respectively, have been previously detected in *An. gambiae* s.s. populations in the country (Martinez-Torres et al. 1998, Weill et al. 2004).

Malaria is also a significant health threat to travelers and military personnel deployed in areas where the disease is endemic. In Côte d'Ivoire, malaria constitutes an important risk in terms of morbidity for expatriates and French troops in rural areas (Migliani et al. 2003) as well as in urban and peri-urban areas, particularly in the city of Abidjan (Eono et al. 1999). The French military camp of Port-Bouët is located on the periphery of the city of Abidjan in a biotope favorable for malaria transmission.

A high level of importance has been accorded to a vector control program to lower the risk of malaria transmission in the camp. In addition to the use of

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deltamethrin-treated nets, permethrin preimpregnated battle dress, and deet repellents, the French army has prioritized outdoor spatial spraying as a key strategy for vector control. Deltamethrin or fenitrothion-based formulations are used, depending on the insecticide's local availability. Application is by vehicle-mounted thermal foggers or hand-carried aerosol generators. Treatment of the entire camp takes 5 wk, and applications are carried out in the mornings throughout the year. Indoor residual spraying is not used routinely for mosquito control. Larval control essentially relies on drainage and the use of temephos is rare. No control is carried out outside the boundaries of the camp.

Notwithstanding these control measures, some authochtonous malaria cases are regularly suspected in the camp. In this article, we present the results of an entomological survey carried out in June 2004 at the military camp of Port-Bouët with the aim of assessing the local malaria transmission risk.

Materials and Methods

Study Site. The city of Abidjan, economic capital of Côte d'Ivoire, has >2.5 million inhabitants. Located on the coast of the Gulf of Guinea, the city is established on a lagoon zone. The climate is tropical, including a long rainy season with a maximum of rainfall in June (500 mm) and a daily average temperature at this time of 25°C. Many Abidjan districts are favorable for the development of anthropophilic mosquitoes because of the presence of numerous available breeding sites, appropriate human hosts and suitable resting places. Many breeding sites are due to poor drainage of waste water, ground water on the surface, and inappropriate attempts at improving sanitation (Zeze et al. 1996). In addition, the lagoon is large and dominates the city, providing many permanent breeding places. The current study was conducted in the French military camp of Port-Bouët, located in southeastern Abidjan, close to the international airport (05.15 N, 03.57 W). The camp covers ≈2 km^2, containing military barracks (permanent and temporary buildings and tents) and shrubby vegetation. The camp is surrounded by a peri-urban environment with modern and traditional houses, vegetable fields, and fallow lands.

Field Mosquito Processing. Mosquitoes were collected in June 2004 on six nights at the time of the new moon. Two standard sampling methods were used: human landing collections and Centers for Disease Control and Prevention (CDC) light trapping. Three catching points were chosen in the camp: the health center, the administrative services (permanent buildings), and the temporary tent camp reserved for combat troops. Landing catches were performed at each catching point on two successive nights from 6:00 p.m. to 7:00 a.m. by a team of eight collectors working in pairs at four sites per point (two sited outdoors in open shelters and two sited indoors in closed tents or permanent buildings). The collectors were rotated every hour between the different sites. They gave prior informed consent and received antimalaria prophylaxis and yellow fever immunization. CDC light traps were placed nightly in the tents used by collectors at rest, and two traps were placed in the workshop inflatable tents on the last night. Traps were run from 6:00 p.m. to 7:00 a.m. each night. Time of sunset and sunrise was 6:30 p.m. and 6:00 a.m., respectively, during the period of study. Individual mosquitoes were labeled according to catching sites, modes and time of capture. They were sorted by genera, and anopheline mosquitoes were identified morphologically by using the keys of Gillies and Coetzee (1987) and the software of Hervy et al. (1998). All mosquitoes were stored individually in numbered vials with desiccant and preserved at −20°C until processing in the laboratory of the Medical Entomology Unit of the Institute for Tropical Medicine, Marseille (France).

Laboratory Mosquito Processing. The head and thorax of anopheline females were tested by enzyme-linked immunosorbent assay (ELISA) for *P. falciparum* circumsporozoite protein (CSP) detection according to Burkot et al. (1984). A random sample of females belonging to the *An. gambiae* complex, together with all CSP-positive anophelines, were identified by polymerase chain reaction (PCR) to species and molecular forms level simultaneously by the method of Fanello et al. (2002). Molecular characterization of the *kdr* and *ace1* mutations were carried out on these mosquitoes as described by Martinez-Torres et al. (1998) and Weill et al. (2004), respectively.

Data Analysis. The human biting rate (HBR) was expressed as the number of female anopheline bites per person per night. The CSP index was calculated as the proportion of mosquitoes found to be positive for CSP. The entomological inoculation rate (EIR) was calculated as the product of the HBR and the CSP index of mosquitoes collected on humans. Conformity of *kdr* and *ace1* frequencies with Hardy–Weinberg expectations was tested using a Pearson chi-square test considered significant when $P < 0.05$. Confidence interval was calculated by the exact method.

Results

Mosquito Collections. In total, 4,776 mosquitoes were caught (54.8% *Culex* spp., 25.6% *Anopheles* spp., 19.0% *Mansonia* spp., and 0.6% *Aedes* spp.) during 24 human-nights and eight CDC light-trapping sessions (Table 1). Three anopheline species were encountered during the period of study: *An. gambiae* s.l., *An. funestus* group, and *An. ziemanni.* The main malaria vector was *An. gambiae* s.l.

Biting Rates and Biting Behavior of *An. gambiae* s.l. Overall, the human landing catches gave an average biting rate of 44.3 *An. gambiae* s.l. bites per person per night; 49.5% of host-seeking *An. gambiae* s.l. were captured indoors, indicating that this species was equally endophagic as exophagic. The distribution of *An. gambiae* s.l. bites by hours is shown in Fig. 1. The peak biting time indoors was between 1:00 and 6:00 a.m., whereas outdoors, biting peaked between 2:00 and 5:00 a.m.

Table 1. Mosquitoes collected in the French military camp of Port-Bouët in June 2004: distribution by species, sampling methods and places of capture (indoor or outdoor)

Species	Human landing catches (24 human-nights)						CDC light traps (8 sessions)
	Indoors		Outdoors		Total		
	n[a]	HBR	*n*	HBR	*n*	HBR	
An. gambiae s.l.	526	43.8	537	44.8	1,063	44.3	156
An. funestus group	1	0.1	0	0.0	1	0.0	0
An. ziemanni	1	0.1	0	0.0	1	0.0	0
Culex spp.	273	22.8	2,095	174.6	2,368	98.7	250
Mansonia spp.	144	12.0	667	55.6	811	33.8	99
Aedes spp.	6	0.5	21	1.8	27	1.1	0
Total	951	79.3	3,320	276.7	4,271	178.0	505

[a] *n* is number of mosquitoes collected.

CSP and EIR. In total, 1,061 *An. gambiae* s.l. collected by human landing catch were processed by ELISA for *P. falciparum* antigen detection. Four specimens were positive. The circumsporozoite index was 0.38% ($CI_{95\%}$ = 0.10–0.96). The one *An. funestus* specimen collected was negative. Considering the *An. gambiae* s.l. human biting rate of 44.3 bites per person per night, the EIR was calculated as 1.2 infective bites per week for a person who did not protect himself during the period of study.

Molecular Identification of *An. gambiae* s.l. One hundred specimens were randomly selected from the pool of 1,061 *An. gambiae* s.l. processed by ELISA. These, together with the four CSP-positive specimens, were identified to species level by rDNA-PCR. One mosquito consistently failed to amplify. One hundred and three mosquitoes were *An. gambiae* s.s. molecular form M.

***kdr* and *ace1* Mutation Frequencies in *An. gambiae*.** We tested the same 104 PCR-identified mosquitoes for the *kdr* and *ace1* mutations. One mosquito consistently failed to amplify for the *kdr* allele, and three mosquitoes failed to amplify for *ace1* mutation. The genotypic frequencies are shown in Table 2. The *kdr* mutation frequency was 0.70 and the *ace1* mutation frequency was 0.15. The population was at Hardy–Weinberg equilibrium for both *kdr* (P = 0.89) and *ace1* (P = 0.69).

Discussion

Mosquito population densities were remarkable considering the intensive vector control program ongoing in the military camp of Port-Bouët. This is particularly true regarding the number of *Culex* and *Mansonia* species caught outdoors and the *An. gambiae* s.l. HBR of 44.3 bites per person per night. *Aedes* population densities were low in our study, but night captures are not appropriate for such diurnal species. The nocturnal mosquito biting rate was intense, and it is clear that personnel living in the military camp were particularly exposed to mosquito bites if they did not protect themselves. This raises obvious questions as to the efficacy of the current vector control program in terms of reduction of mosquito densities.

Three *Anopheles* species were collected in the camp: one *An. ziemanni*, one *An. funestus*, and a total of 1,219 specimens belonging to the *An. gambiae* complex. *An. ziemanni* is usually an exophagic and zoophilic species and is not considered as a malaria vector (Gillies and Coetzee 1987). *An. funestus* probably does not contribute to the malaria transmission in the camp

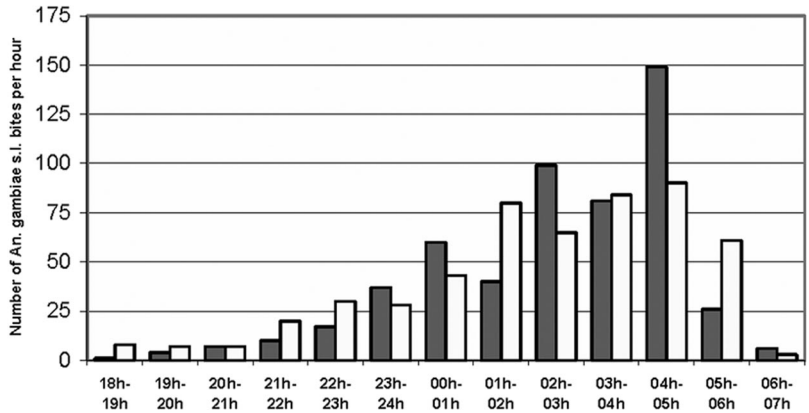


Fig. 1. Distribution by hours of *An. gambiae* s.l. bites indoors (white bars) and outdoors (gray bars) in the French military camp of Port-Bouët in June 2004.

Table 2. Genotypic frequencies for kdr and ace1 loci of a sample of *An. gambiae* collected in the French military camp of Port-Bouët in June 2004.

	kdr	ace1
RR	52 (0.50)	1 (0.01)
RS	41 (0.40)	28 (0.28)
SS	10 (0.10)	72 (0.71)
Total	103	101

RR, resistant homozygous; RS, heterozygous; SS, sensible homozygous. Numbers within parentheses are genotypic frequencies.

in June, but its presence suggests that further work will be necessary to determine its local abundance throughout the year and its potential role in malaria transmission. Only one species of the *An. gambiae* complex was present: *An. gambiae* s.s. The absence of *An. arabiensis* is consistent with its distribution restricted to the northern savannah areas of Côte d'Ivoire (Gillies and Coetzee 1987). The absence of *An. melas* is more unexpected considering previous studies describing this species from the southern forested areas of Côte d'Ivoire (Hamon et al. 1956, Doucet et al. 1960, Coz 1973) and particularly in Abidjan (Binson and Doucet 1957). Abundant rainfall occurs in June, reducing salinity of breeding places and might explain this situation. However, these results are consistent with previous work showing that *An. gambiae* s.s. was the most abundant and widespread malaria vector in the city and more recent work of Chandre et al. (1999a) who detected only *An. gambiae* s.s. in Abidjan.

Only the M molecular form of *An. gambiae* was detected in our study. A correlation between M and S forms and ecological factors has been shown, the M form being predominant in forest zones and the S form in savannah zones, although the two forms are found in sympatry at many sites (Della Torre et al. 2001). Furthermore, the M form of *An. gambiae* is known to be associated with flooded or irrigated sites typified by extensive rice cultivation or other permanent breeding conditions, whereas the S form is characteristically found in rain-dependent temporary breeding sites (Touré et al. 1998). The identification of only the M form in the camp of Port-Bouët is indicative of the presence of permanent breeding places in the area. The irrigated vegetable fields located near the military camp are probably a good source for *An. gambiae* breeding. Indeed, human activity in urban and peri-urban areas usually renders the situation favorable for the breeding of most malaria vectors in sub-Saharan Africa (Robert et al. 2003, Wang et al. 2005).

The human night biting pattern is interesting in terms of individual personal protection. Our results indicate that in June, *An. gambiae* is preferentially biting in the second part of the night, when personnel are usually sleeping. Nevertheless, some biting activity takes place in the early part of night (from 6:00 p.m.) when personnel are often outside and continue to bite early in the morning (up to 7:00 a.m.) when personnel are generally no longer protected against mosquito bites. The use of skin repellents is not usual early in the morning.

The EIR calculated in our study is for June, which is usually the wettest month in Abidjan. At this time, a person who did not take any precautionary measures would receive more than one infective bite per week. In endemic areas, the average EIR in peri-urban environments is estimated as 48.5 infective bites per person per year (Robert et al. 2003). Our data are consistent with this estimation. Malaria transmission in the military camp of Port-Bouët, therefore, has been clearly demonstrated for the first time, confirming the suspected occurrence of local malaria transmission in the camp for a few years. As a result, the efficacy of the current vector control program, in terms of malaria transmission reduction, is questionable.

Previous studies indicated that *An. gambiae* s.s. populations from Abidjan were susceptible to pyrethroids (Chandre et al. 1999a) and that the *kdr* mutation was not present (Martinez-Torres et al. 1998). Subsequently, studies have shown a rapid appearance of the phenomenon with cross-resistance to many pyrethroids and a frequency of the *kdr* mutation of 0.39 (Chandre et al. 1999b). Our data showed a *kdr* frequency of 0.70 in the M molecular form of *An. gambiae* from Port-Bouët, which contrasts with Della Torre et al. (2001) who observed the *kdr* mutation only in the S form and not at all in the M form in Côte d'Ivoire. The *kdr* mutation was initially found only in the S form in West Africa but more recently has been detected in the M form in Benin (Weill et al. 2000), Burkina Faso (Diabate et al. 2003), and Ghana (Yawson et al. 2004). However, we here present the first report of the *kdr* mutation in the M form of *An. gambiae* from Abidjan.

Resistance to carbamate and organophosphate insecticides is also widespread in West Africa, including Côte d'Ivoire. N'Guessan et al. (2003) and Corbel et al. (2003) revealed the presence of an insensitive acetylcholinesterase in populations of *An. gambiae* from Bouaké and central Côte d'Ivoire by biochemical assays. Weill et al. (2004) detected the *ace1* mutation in both M and S forms from Bouaké. Here, we present the first report of this mutation in the M form of *An. gambiae* from Abidjan.

The data presented here represent a first step in documenting the current state of insecticide resistance in the M form of *An. gambiae* in the military camp of Port-Bouët. The significance, in terms of physiological resistance, of the occurrence of the *kdr* and *ace1* mutations in *An. gambiae* M form is far from clear as shown by Chandre et al. (1999b, 2000) who measured lower correlations between *kdr* frequencies and resistance levels to pyrethroids. Other mechanisms of insecticide resistance apparently exist as shown by Awolola et al. (2003) who did not find the *kdr* mutation in specimens of *An. gambiae* M form from Nigeria resistant to pyrethroids. As a result, insecticide susceptibility tests should be carried out to assess physiological resistance levels in *An. gambiae* in Port-Bouët.

In terms of effective vector control at the military camp, the current strategy needs urgent reassessment.

There is evidence from malaria transmission studies in West Africa indicating that the presence of the *kdr* mutation does not reduce the efficacy of insecticide-treated bed-nets (Chandre et al. 2000, Darriet et al. 2000, Corbel et al. 2004). Deltamethrin-impregnated bed-nets can therefore continue to be used systematically by the personnel living in the camp. Not surprisingly, space spraying of insecticides obviously had little or no impact on *An. gambiae,* and application of residual insecticides inside the buildings or tents should be considered to improve the malaria vector control program. Choice of insecticide should depend on the results of susceptibility tests on the vector species, including *An. funestus* should this species be demonstrated to be a significant vector at other times of the year. The rotation of different classes of insecticides might be a potential tool for resistance management in the future.

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